

Vellacombe Discovery Ltd

Indazoles: N-substitution map and positional assignments

Route scout and delivery — final report, revision A

Programme: VD-15-134

Accession: VA20151235

Client: Feldorune Therapeutics BV

Date: 31 Mar 2016

Author: Dr R. Calderwood, Head of Medicinal Chemistry

1 Final material schedule

Feldorune Therapeutics BV received 40 compounds under VD-15-134. Analogue preparation began on 28 Dec 2015; the interim checkpoint was 12 Feb 2016, and chemistry completion was recorded on 30 Mar 2016. This final Route scout and delivery report is issued by Dr R. Calderwood on 31 Mar 2016.

The analogue set was delivered with positional assignments, allowing the client to distinguish divergent potency trends. Position belongs to the chemical identity, not to an annotation added after a biological result becomes interesting. Hassan Darwish maintained the route relationship to the N-substituted product; Ruth Calderwood treated positionally different products as distinct comparisons. The final chemistry conclusion does not rank receptor-associated kinase activity.

Table 1 brings the identifier, assigned structure, molecular weight, isolated yield, UV area purity and amount into the same entry. Only this schedule constitutes the reported delivery. Suggested structures elsewhere in the report are not additional isolated compounds.

Table 1. Compounds delivered; final isolated yield, HPLC purity in UV area percent at 254 nm and amount in mg.

Compound	SMILES	MW	Yield (%)	HPLC purity (UV area %, 254 nm)	Amount (mg)
FT09-07010	<chem>CN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(C#N)cc4)cc23)CC1</chem>	374.4	26	99.3	26
FT09-07012	<chem>N#Cc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCOCC4)c3c2)cc1</chem>	361.4	27	98.8	27
FT09-07018	<chem>N#Cc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCC(N)CC4)c3c2)cc1</chem>	374.4	55	97.1	35
FT09-07028	<chem>CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4cc(F)cc(F)c4)cc23)cc1</chem>	420.4	88	97.3	17

	23)cc1				
FT09-07030	CN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccnc4)cc23)CC1	350.4	46	96.3	41
FT09-07036	Nc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCC(N)CC4)c3c2)cc1	364.5	23	94.1	35
FT09-07042	O=C(NCN1CCOCC1)c1n[nH]c2cc(-c3cncnc3)cc12	338.4	33	96.5	17
FT09-07046	O=C(NCN1CCC(C1)c1n[nH]c2ccc(-c3ccnc3)cc12	321.4	31	97.5	30
FT09-07052	CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4cncnc4)cc23)cc1	386.4	75	95.5	36
FT09-07054	O=C(NCN1CCC(O)CC1)c1n[nH]c2ccc(-c3cncnc3)cc12	352.4	67	95.8	31
FT09-07058	NC1CCN(CNC(=O)c2n[nH]c3ccc(-c4cncnc4)cc23)CC1	351.4	43	93.5	52
FT09-07060	Nc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCOCC4)c3c2)cc1	351.4	58	98.9	45
FT09-07070	CN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(N)cc4)cc23)CC1	364.5	41	95.3	39
FT09-07072	O=C(NCN1CCC(C1)c1n[nH]c2ccc(-c3ccc(F)cc3)cc12	338.4	4	95.4	8
FT09-07076	CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4ccncc4)cc23)cc1	385.4	38	98.3	33
FT09-07078	CCN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccnc4)cc23)CC1	364.5	50	96.9	38
FT09-07084	N#Cc1ccc(-c2ccc3[nH]nc(C(=O)NCc4ccncc4)c3c2)cc1	353.4	27	96.1	41
FT09-07088	O=C(NCN1CCC(C1)c1n[nH]c2ccc(-c3ccncc3)cc12	321.4	36	95.7	20
FT09-07090	CN1CCN(CNC(=O)c2n[nH]c3ccc(-c4cc(F)cc(F)c4)cc23)CC1	385.4	64	98.7	31
FT09-07094	CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4ccnc4)cc23)cc1	385.4	58	94.5	43
FT09-07100	COc1ccc(-	358.4	45	95.0	5

	<chem>c2ccc3[nH]nc(C(=O)NCc4ccncc4)c3c2)cc1</chem>				
FT09-07102	<chem>NC1CCN(CNC(=O)c2n[nH]c3ccc(-c4cc(F)cc(F)c4)cc23)CC1</chem>	385.4	40	98.5	24
FT09-07106	<chem>O=C(NCN1CCOCC1)c1n[nH]c2cc c(-c3ccc(F)cc3)cc12</chem>	354.4	58	97.6	33
FT09-07108	<chem>NC1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(F)cc4)cc23)CC1</chem>	367.4	41	97.2	48
FT09-07114	<chem>COc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCN(C)CC4)c3c2)cc1</chem>	379.5	70	97.1	51
FT09-07118	<chem>O=C(NCN1CCOCC1)c1n[nH]c2cc c(-c3cc(F)cc(F)c3)cc12</chem>	372.4	44	97.0	5
FT09-07120	<chem>O=C(NCN1CCC(O)CC1)c1n[nH]c2ccc(-c3cccnc3)cc12</chem>	351.4	36	96.6	25
FT09-07124	<chem>O=C(NCN1CCC(C1)c1n[nH]c2ccc(-c3cc(F)cc(F)c3)cc12</chem>	356.4	61	98.6	19
FT09-07132	<chem>Nc1ccc(-c2ccc3[nH]nc(C(=O)NCc4ccncc4)c3c2)cc1</chem>	343.4	57	95.1	37
FT09-07144	<chem>CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4ccc(C#N)cc4)cc23)cc1</chem>	409.4	49	97.3	11
FT09-07148	<chem>O=C(NCc1ccncc1)c1n[nH]c2ccc(-c3ccncc3)cc12</chem>	329.4	54	97.2	27
FT09-07150	<chem>CC(=O)Nc1ccc(CNC(=O)c2n[nH]c3ccc(-c4ccc(N)cc4)cc23)cc1</chem>	399.5	35	94.5	28
FT09-07156	<chem>COc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCC(O)CC4)c3c2)cc1</chem>	380.4	42	96.4	14
FT09-07160	<chem>NC1CCN(CNC(=O)c2n[nH]c3ccc(-c4cccnc4)cc23)CC1</chem>	350.4	29	96.3	49
FT09-07162	<chem>CCN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(N)cc4)cc23)CC1</chem>	378.5	40	96.2	17
FT09-07168	<chem>N#Cc1ccc(-c2ccc3[nH]nc(C(=O)NCN4CCCC4)</chem>	345.4	39	96.4	12

	c3c2)cc1				
FT09-07172	CCN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(C#N)cc4)cc23)CC1	388.5	51	97.0	29
FT09-07178	CCN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(F)cc4)cc23)CC1	381.5	43	98.2	31
FT09-07180	O=C(NCc1ccncc1)c1n[nH]c2ccc(-c3cncnc3)cc12	330.4	14	96.0	45
FT09-07186	CCN1CCN(CNC(=O)c2n[nH]c3ccc(-c4ccc(OC)cc4)cc23)CC1	393.5	32	97.3	28

2 Remit, chronology and responsibility

The remit was Analogue preparation for the FT09 set. The report covers route decisions affecting access to material and the identity basis needed for the client comparison. The programme turnaround was 93 days. That figure describes the programme interval, not a reaction time or assay incubation.

Dr R. Calderwood led the programme, Dr H. Darwish directed synthetic execution and Clara Fenwick led analytical release. The final gate closed on 30 Mar 2016; this report follows completion rather than reopening the route as a draft. Programme accession VA20151235 links the material schedule to the benchmark below.

Earlier intermediates remain useful experimental context but are not counted again as delivered compounds. A repeat request should name the released identifier and the chemistry question. The final schedule is the endpoint for reconciling the set; individual compound cards remain the source for recovering a preparation.

Table 2. Programme benchmark and administrative reconciliation.

Metric	Value
Project	VD-15-134
Client	Feldorune Therapeutics BV
Compounds tested	40
Series / chemotype	Feldorune indazole series 09 (Indazole)
Assays	Custom synthesis and analogue preparation
Turnaround (calendar days)	93
Price	GBP 42,300

3 Chemistry hypothesis

The indazole programme required an N-substitution map with the position explicit. Feldorune Therapeutics BV needed to distinguish divergent potency trends; a generic N-substituted label could hide the chemical difference needed for interpretation. The working hypothesis was that positional assignment must travel with the isolated compound throughout preparation and delivery. The task was to supply that distinction reliably, not to select a potency trend without biological measurements in this report.

The programme was organised around material needed to answer the client question. Route success was judged at the isolated product, not at an attractive intermediate or plausible scheme. This distinction limits the final claims: chemistry establishes access and identity for the delivered set but supplies no biological ranking that has not been measured here. The proposed follow-up must retain the same distinction.

4 Route gate and sequence decision

The analogue set was delivered with positional assignments, allowing the client to distinguish divergent potency trends. Position belongs to the chemical identity, not to an annotation added after a biological result becomes interesting. Hassan Darwish maintained the route relationship to the N-substituted product; Ruth Calderwood treated positionally different products as distinct comparisons. The final chemistry conclusion does not rank receptor-associated kinase activity.

The interim discussion on 12 Feb 2016 was a route checkpoint, not the delivery endpoint. By 30 Mar 2016, the completed material schedule could be reconciled with the programme. A repeat chemist should preserve the reason for the route choice as well as the sequence. Repeating an operation without knowing which recovery or identity risk it controlled can recreate the original difficulty.

5 Analogue preparation

N-substitution can produce alternatives sharing formula and nominal mass. Analogue preparation must preserve the link between substitution operation, isolated fraction and position assigned in the compound card. The final drawing is useful for the client, but the route history lets another chemist recover the same material. An unmade positional proposal cannot be entered as a released member merely because a related alternative was isolated.

For repeat work, follow the compound card for the chosen identifier and reconcile the intended product with Table 1. A common scaffold is not a common experimental outcome. Substitution changes can affect work-up, purification and the evidence needed for assignment. No unprepared variant is claimed to have crossed the route gate merely because its drawing resembles a delivered compound.

6 Separation, work-up and recovery

Purification separates material; assignment identifies what has been separated. Similar chromatographic behaviour can make a narrow collection window insufficient evidence of positional identity. Table 1 reports the final compounds with their assigned structures and recoveries. Its yields are not an N-substitution product ratio, and UV area purity must not be repurposed as a measure of positional selectivity. The distinction remains important when comparing recovery across the set.

Isolated yield is a route measure and delivered amount is a material measure. Both matter because the comparison requires the actual compound in hand. Neither column tells a chemist where loss occurred. Return to intermediate and fraction entries for that question. An estimate derived only from the final table would conceal the difference between reaction performance and recovery.

7 Identity and assignment

Clara Fenwick handled release with positional assignment retained as an identity requirement. Formula and mass cannot distinguish alternatives of the same composition. The substitution position must stay explicit when a compound is transferred to a biology entry or selected for repeat synthesis. Potency is not an assignment method: an isolate belongs to its chemical position whether its eventual response follows the expected trend or departs from it.

The release entry must stay connected to the isolate. A structure redrawn for a client presentation should retain the chemistry identifier so the assignment basis can be recovered. This report does not reproduce raw spectra or supply invented peak assignments. It records the programme conclusion and explains where the underlying experimental history is retained.

8 Purity statement and analytical limits

The HPLC values in Table 1 are UV area percent at 254 nm. They describe the reported chromatographic profile under that detection basis. They are not weight percent, solvent content or a direct assay of active parent in a weighed portion. The amount field is expressed in mg and must remain separate from purity when planning the comparison.

Clara Fenwick led analytical release. Identity and chromatographic purity answer complementary questions: a dominant peak cannot independently establish the assigned connection, position or configuration. An assigned structure also does not remove the need to consider impurities. The schedule retains the available numerical values without adding a universal purity threshold or an unstated correction to the delivered amounts.

A client query concerning precipitation, unexpected peaks or identity should return against the delivery identifier. The compact table is a reconciliation aid, not a substitute for the analytical entry needed to investigate a particular sample.

9 Isolated yield and material accounting

Final yields and amounts remain separated by compound. No pooled recovery or campaign mean has been calculated. Such a figure could blur route differences and imply a common experimental basis that the delivery table does not establish. The practical question is whether identifier, structure and amount refer to the same released material.

Compound cards may contain intermediate masses, recovered fractions and material held for further work. Those entries do not automatically belong to the client delivery. Use Table 1 for the delivered set and the route scout for an individual recovery question. A request for more material must distinguish repeat product from a new proposed analogue.

The programme closed with 40 delivered compounds. This count is not enlarged by an intermediate, an alternative drawing or a proposed comparison. Detailed weighing and transfer history belong to the individual chemistry entries; the final schedule supplies report-level reconciliation.

10 Client hand-off and identifier continuity

The FT09 identifiers connect isolated chemistry to the client comparison. Use VD-15-134 and accession VA20151235 for clarification. Assigned structures help reconcile a vial label with its intended product, but the delivery identifier remains the primary link to prepared material.

If a client transfers material for biological preparation, retain the original identifier and distinguish that later preparation description from chemistry release. Replacing an identifier with a proposed structure name risks merging different samples under a general scaffold label. The risk matters particularly where apparently similar structures address different chemical questions.

Chemistry completion was recorded on 30 Mar 2016. This report supplies no unstated courier date, receipt signature or storage condition. Dr H. Darwish is the route contact; Clara Fenwick is the analytical reconciliation contact; Dr R. Calderwood retains responsibility for the overall chemistry conclusion.

11 Interpretation for the client comparison

Feldorune Therapeutics BV can distinguish divergent potency trends because the delivery preserves positional identity. No numerical potency measurements or trend ranking are included here. Keep substitution position as an explicit descriptor and return later results against the delivery identifier. Grouping compounds only by side-chain name would lose the chemical distinction that motivated the programme. Reconcile the position and tested material before interpreting any trend.

The useful follow-up is a comparison whose members can be traced to released material. Keep the intended chemical difference explicit. Do not combine a proposed product and an isolated compound within a result entry. A later biological trend may justify new chemistry, but cannot change retrospectively which material was delivered.

This is a chemistry close-out. It provides no invented response values, selectivity ratios or potency rankings. Programme turnaround is not an assay setting, and isolated yield is not an indicator of target activity.

12 Next analogue and carried route risk

The proposed next comparison holds peripheral substitution constant while examining the alternative N-substitution position. It is not an additional delivery or a claim of matched potency. The principal risk is copying an assignment from a neighbouring compound rather than establishing it for the new isolate. Repeat preparation should preserve the naming basis and pass the same identity gate before biological interpretation. An assigned class should not be discarded solely because another appears more promising.

A repeat route scout should identify the operation controlling the original risk and the evidence that closed it. Do not omit that gate because the final compound looks familiar. Substituent changes and material history can affect recovery without changing the intended chemical question. Preserve the link between the proposed route and the actual isolated outcome.

The report does not qualify a working scale beyond the delivered chemistry or impose new process conditions. Recover charge quantities, timings and purification parameters from the relevant experimental entry. A change in the established sequence needs its own reason and compound-card history.

13 Follow-up decision and status

The next analogue described above is proposed chemistry. It has no additional delivery entry, assigned yield, purity, amount or biological result in this report. Its purpose is a defined extension of the successful route while retaining the chemical distinction that motivated the programme.

Dr R. Calderwood recommends that Feldorune Therapeutics BV reconcile the intended comparison with the released identifiers before expanding preparation. Once the chemical question is selected, the next route scout can be scoped around it. A broad set assembled before that choice may add material without clarifying interpretation.

The gate for a new analogue remains identifiable isolated material. A convincing scheme is not a completed delivery. This final report closes the present programme and proposes a reasoned next step without presenting it as work already performed.

Appendix A — Delivery-table conventions

Compound is the delivery identifier linking material to chemistry. SMILES is the compact assigned-structure field. MW is the reported molecular-weight field associated with it. These fields must not be used to infer an unstated counter-ion, solvate or assignment detail. The drawing summarises identity; the compound card preserves its basis.

Yield (%) is the reported isolated yield. HPLC purity is UV area percent at 254 nm. Amount (mg) is the reported isolated amount. Keep the bases separate. The table supplies no raw peak areas, moisture correction, uncertainty estimate or across-programme normalisation. It also does not substitute for a preparation-specific experimental entry.

To reproduce a delivered product, select the identifier, recover its compound card and reconcile the final isolate with its route history. To extend the chemistry, create a route proposal rather than repurposing an existing delivery entry for an unmade structure.

Appendix B — Experimental history and repeat instructions

Compound cards are the primary experimental link for this programme. Consult the relevant card for the N-substitution operation, fraction history and positional assignment basis. The final report gives the set-level delivery schedule and route conclusion. It does not invent spectral shifts or imply that a compact structure field contains the full assignment evidence. Repeat preparation must recover the card for the intended position and reconcile the final isolate before release.

A repeat request should identify the compound, why additional material is needed and the route question to preserve. The chemist can then recover the operation sequence and compare the repeat with the established identity basis. A broad scaffold description is insufficient when the conclusion depends on separation, selective access or assignment.

The report remains the final account of delivered chemistry. It does not overwrite experimental entries or create replacement conditions. If subsequent preparation departs from the established route, retain that departure in its own compound card and distinguish the new material history from the present delivery.

14 Final conclusion and authorisation

The analogue set was delivered with positional assignments, allowing the client to distinguish divergent potency trends. Position belongs to the chemical identity, not to an annotation added after a biological result becomes interesting. Hassan Darwish maintained the route relationship to the N-substituted product; Ruth Calderwood treated positionally different products as distinct comparisons. The final chemistry conclusion does not rank receptor-associated kinase activity.

The final delivery comprises 40 isolated compounds for Feldorune Therapeutics BV. Amounts, isolated yields and UV area purity are recorded in Table 1. The route supports the reported client comparison; proposed follow-up structures remain outside the delivery. No biological ranking is assigned in this conclusion.

Issued on 31 Mar 2016, after completion on 30 Mar 2016. Dr R. Calderwood authorises the chemistry conclusion. Dr H. Darwish is identified for synthetic execution and Clara Fenwick for analytical release. Revision A is final and has no open comments or provisional replacements for the material schedule.

Name	Role	Signature	Date
Dr R. Calderwood	Head of Medicinal Chemistry — final report authorisation	<u>D Calderwood</u>	31 Mar 2016